NORTHERN ILLINOIS UNIVERSITY

Housing Burden and Insurance Expenditures

A Capstone Submitted to the

University Honors Program

In Partial Fulfillment of the

Requirements of the Baccalaureate Degree

With Honors

Department Of

Economics

By

Melanie Lytell

DeKalb, Illinois

May 9th, 2026

Abstract

Housing costs shape the way families spend on protection and security. This study looks at how the share of income devoted to housing, the housing burden, influences household spending on four types of insurance: vehicle, life, personal, and health. Using data from nearly 92,000 households and regression models that account for state differences, I find that even small increases in housing burden shift insurance choices in meaningful ways. A one-percentage-point rise in housing burden is linked to a 2.1% increase in vehicle insurance spending and a 0.5% increase in life insurance, but a 0.7% decrease in personal insurance and a 0.5% decrease in health insurance. Age, education, marital status, and housing tenure also matter; older households spend more on health and life insurance, while renters and student housing residents consistently spend less across all categories. Higher education levels are tied to greater spending, especially for personal and health insurance. Patterns in related expenses, such as alcohol, medical services, and vehicle costs, reinforce these findings. Taken together, the results show how housing affordability pressures impact household budgets, encouraging spending in some areas while crowding out others. These insights highlight the trade-offs families face and underscore the importance of housing policy in shaping financial security and access to protection.

Keywords: Housing burden, Insurance expenditures, Vehicle insurance, Life insurance, Personal insurance, Health insurance

Housing Burden and Insurance Expenditures

Introduction

This study investigates the circumstances that impact household expenditures on vehicle, life, personal, and health insurance. Detailed regression models are used to analyze how demographic characteristics (age, race, marital status, and education), household dynamics (family size and housing tenure), income, and state that the household is in affect how much households spend on each insurance. The dependent variable for each model is a continuous variable that measures how much each household spends on each insurance, which allows me to capture if households purchase insurance or not, and how much they are investing into protection.

Rising housing costs have placed growing financial pressure on households, increasing concerns about how these higher housing burdens affect participation in insurance. As rent and mortgage payments take up more of households' income, they may be forced to make trade-offs that compromise their protection. Ongoing disparities in insurance access across income and race emphasizes the disproportionate distribution of risk protection in the United States. The gaps shape who can have coverage; the quality of available policies, and the extent that households can rely on insurance to protect them from unexpected events. Insurance spending is a key indicator of risk management. It reflects the accessibility of insurance coverage and the perceived value of protecting against certain risks, such as damage to property and medical emergencies. In the United States, insurance markets are a combination of private insurance, public programs, and employer-sponsored coverage, indicating that household decisions are impacted by personal circumstances that are related to the person's job or lack thereof to some

extent. Understanding how factors impact the demand for insurance is critical due to ongoing disparities in insurance access across income, race, and location.

This topic is important because insurance is more than a financial asset; it is also a safety net. When households don't invest enough into insurance, they become more liable to economic shocks, health emergencies, and long-term instability. My analysis further contributes to the existing literature on financial behavior, risk preferences, and financial protection. It also provides insights for government officials and insurance companies that are trying to close the gaps in coverage and make more equitable insurance systems. By modeling expenditures across multiple types of insurance, this is a comprehensive look of how American households assess their risk.

Literature Review

Prior literature highlights several drivers behind participation in insurance markets: while some studies find limited evidence of adverse selection, others show that risk aversion, income constraints, and behavioral biases significantly influence insurance participation. Prior research also indicates that housing affordability can shift demand from private insurance toward public insurance like Medicaid. Adding to these findings, my study investigates how housing burdens shape household decisions across multiple types of insurance, aiming to clarify what drives underinsurance in financially strained populations.

Asymmetric Information

Cardon and Hendel (2001) look at the presence of asymmetric information in the health insurance industry. Their hypothesis is that if there is adverse selection present, there should be a significant connection between unobservable private information, insurance choices, and

healthcare consumption. For their analysis, they use the National Medical Expenditure Survey from the Agency of Health Care Policy Research data from 1987. In the models, they use variables such as demographics, income, insurance information, and health care consumption. A discrete continuous model is used to integrate health insurance and healthcare demand. The model has a two-stage decision process: stage 1 is when consumers choose their insurance policy based on expected utility, given they know about their health risk, and stage 2 is when consumers decide their health care consumption after they learn about their health state. A method of moments estimator is used to link predicted behavior and observed behavior, and a Hausman specification test is performed to see if there is independence between insurance status and the error terms. The study finds that adverse selection is not a significant problem in the health insurance market for the single working-age population and the difference in healthcare expenditures between the insured and uninsured is not due to lack of information, but price sensitivity. Income is a significant predictor of insurance status and health care consumption, with insured individuals earning on average more than twice the uninsured ($22,059 vs $10,632).

Building on the theme of asymmetric information, Cohen (2005) looks at asymmetric information in the automobile insurance market. The hypothesis is that higher-risk individuals are more likely to purchase more insurance coverage. The data used in the study comes from an automobile insurer operating in Israel from 1995-1999 and the variables used in the models are: demographics, car characteristics (engine size, model year, car value, usage type), income, driving characteristics (years of experience, past claims, history, number of drivers), policy details (deductible choice, premium, claims submitted, and total costs) and self-reported vs observed claims. Several estimation methods are used to test the hypothesis, starting with

Ordinary Least Squares to test the correlation between deductible choice and the number of claims. A poisson regression is used to model the number of claims that exceeded the deductible; a bivariate probit regression is used to estimate the correlation between deductible choice and the frequency of claims; a mean comparison test is used to compare self-reported claims by new customers with observed claims and a logit regression was used to analyze the likelihood of policyholders leaving the insurance because of history. The study finds that new customers that chose lower deductibles were significantly more likely to file claims, and the coverage-accidents correlation is statistically significant for policyholders with 3+ years of experience and new customers are also more likely to underreport the number of claims when changing insurers while the coverage and accidents correlation support the presence of adverse selection. Individuals with higher income were more likely to choose lower deductibles.

Chiappori and Salanie (2000) test if there is asymmetric information in the French automobile insurance market with the hypothesis is that if asymmetric information is present, there should be a positive correlation between the level of insurance coverage and their likelihood of getting into an accident. The French Federation of Insurers data is used since it represents 21 companies which manages 70% of the automobile insurance market in France. The data is from 1989, with an emphasis on drivers who obtained their licenses between 1986 and 1988. Three statistical tests are done in this study to test their hypothesis: two probit regressions, a bivariate regression, and a chi-squared independence test. The dependent variable in the first probit regression is a binary variable that looks at the type of insurance coverage a person has where comprehensive = 1, and legal minimum = 0. The dependent variable in the second probit regression is a binary variable that indicates whether the individual has reported an at fault accident where at least one = 1, and none or no fault = 0. The independent variables are the same

for each model: demographics (sex, age of driver, profession), vehicle characteristics (make, size, age of car), and use area (binary variable that measures whether the car is used in a city). These regressions are estimated separately under the assumption of conditional independence of errors and the purpose of having the two separate regressions is to examine the relationship between the overlooked variables by analyzing the residuals indirectly. In the case of the bivariate probit model, both coverage choice and accident occurrence are dependent variables, and the independent variables are the same as the separate probit models, but there is a focus on drivers with no recent driving history, and the purpose of using the bivariate probit model is to test whether the errors are correlated. If so, there is evidence of asymmetric information. For the chi-squared independence test the dependent variables are coverage choice and accident occurrence and the independent variables are a set of six binary variables (sex, car make, car size, driver age, car age) which are used to create cells for testing independence and the purpose of this test is to test independence nonparametrically without error distributions. Nonparametric chi-squared tests were used to test independence, and a Kolmogorov-Smirnov test was used to compare the distribution of the test statistics with the theoretical chi-squared distribution. Through all these tests, there was no evidence of asymmetric information, no evidence of risk type revealed through experience, and no evidence of moral hazard. Income helped to explain who chose more coverage but does not drive the results.

When put together, these studies give mixed results on if there is asymmetric information in insurance markets. Cardon and Hendel (2001) and Chiappori and Salanie (2000) both do not support adverse selection being in insurance markets, which suggests that insurance choices are driven by other factors like price sensitivity. On the other hand, Cohen (2005) found statistically

significant evidence of adverse selection from more experienced drivers, which suggests that people with greater risk choose to have better coverage.

For the broader question of how housing burdens impact insurance participation, these findings suggest that affordability and price sensitivity are more influential than having private information. This emphasizes the point that housing costs and burdens can play a really critical role in household insurance decisions. It reinforces the need to look at the financial conditions of a household along with informational asymmetry.

Risk Aversion and Demand for Insurance

Cleeton and Zellner (1993) look at the relationship between income, risk aversion, and the demand for insurance. They hypothesize that insurance demand is influenced by the income of the consumer, the degree of risk aversion, and the relationship between income and the size of potential losses. The analysis used in this paper is theoretical using frameworks like the Ehrlich-Becker framework to derive comparative static results. The included variables are income, potential loss, price of insurance, risk aversion (degree), and the probability of loss. There were many relevant findings. At actuarially fair prices, a risk-averse person will completely insure against a potential loss and if the likelihood of loss increases along with income, the demand for insurance increases proportionally, which makes insurance a normal good. At non-actuarially fair prices under decreasing absolute risk aversion, an increase in income increases demand, which makes insurance a normal good. At non-actuarially fair prices under increasing absolute risk aversion, an increase in income reduces demand, which makes insurance an inferior good. The income elasticity of demand for insurance is equivalent to the income elasticity of the potential loss at actuarially fair prices. When insurance is priced fairly, demand for insurance

rises or falls simultaneously with the size of potential losses. In this case, the way insurance demand responds to income is the same as how potential losses scale with income, since insurance simply converts risks into payments. This means insurance is not a normal good in itself; its demand grows only as risk exposure grows with income, rather than reflecting independent preferences for insurance.

Expanding the relationship between risk preferences and insurance behavior, Outreville (2014) looks at the relationship between risk aversion, risk behavior, and demand for insurance using demographic factors as variables. This paper hypothesizes that risk aversion has a significant impact on insurance demand, and several data sources are used: surveys, international studies, experimental data, insurance-specific datasets. Many variables were used: socio-demographic, economic, behavioral, and cultural dimensions, and the data includes studies from the 1950's to the 2000's with specific studies focusing on periods like 1962-1963, 1980-1993, and 1992-2001. Several estimation methods were used, such as regressions, were used to estimate relationships between risk aversion and socio-economic/demographic variables, tobit models for censored data (insurance purchases), cross-sectional and panel data regressions were used for macroeconomic studies, lab experiments were performed to measure risk preferences, and game shows were analyzed to see how participants' risk attitudes are shaped, and questionnaires were used to assess individual risk preferences. The author integrates results from many microeconomic and macroeconomic studies by organizing them by comparing how demographic variables influence demand and risk aversion across contexts. The author finds that risk aversion is positively correlated with insurance consumption, particularly with life insurance. Women are more risk averse than men, ultimately leading to increased demand, and risk aversion increases with age, especially for those above the age of 65, higher education

correlates with lower-risk aversion, but there is a greater awareness of the necessity of insurance.

These studies highlight the role of risk aversion in shaping demand for insurance, but they approach the topic from different angles, Cleeton and Zellner (1993) focus on a theoretical model that emphasizes how income and pricing interact with risk preferences to influence insurance decisions. In contrast, Outreville (2014) provides empirical evidence showing how demographic and behavioral factors correlation with risk aversion and then with insurance participation, Outreville's findings suggest that these real-world behaviors often imply that there are deeper social and psychological patterns.

For households with higher housing burdens, these insights are particularly relevant. If risk aversion increases with age or vulnerability, greater demand for insurance is expected. On the other hand, when housing costs are a large share of income, even risk averse individuals may not decide to have insurance because they cannot afford it. This underscores the importance of looking at financial factors as well as behavioral factors when analyzing insurance participation in high housing burden households.

Changes in Demand

Stutz and Kartman (1982) looks at housing affordability and price variations in the United States. This study hypothesizes that housing price variations across cities are influenced by in-migration, income growth, shifts in regional demand, vacancy rates, constraints on the supply of land, labor and capital, and environmental restrictions. The data used in this study came from multiple sources like the U.S. Bureau of the Census, the Federal Home Loan Bank Board, NOAA and others. The dependent variable is the average housing price (transaction

price) for January-February 1981 across 30 U.S. cities. The independent variables are solar radiation (used for regional demand), mean temperature difference (used for regional demand), per capita income, vacancy rates, loan rates, total population, suburban population growth, industrial utility rates, and union wage rates. The data is from 1981, and some variables reference earlier years. A stepwise regression is used to identify the order of entry of variables into the model and to see their incremental contribution to the explanation of housing price variation. After that, a standard multiple linear regression is used. Regional demand factors are the most significant determinants of housing price levels. Supply-side factors like loan rates and labor costs were less influential.

Building on the theme of housing affordability, Cao, et al. (2024) look at how health insurance coverage changes under the Affordable Care Act among high housing costs. Their hypothesis is that the Affordable Care Act has a more significant impact on households with higher housing burdens compared to lower burdened households. The source of their data is the American Community Survey and the variables included in the models are rent-to-income ratio, health insurance coverage types, and state-level ACA implementation details and ranges from 2010-2018. A difference-in-difference analysis is used to compare insurance coverage trends between expansion and non-expansion states before and after Medicaid expansion, which happened in 2014, event study analysis is used to test the parallel trend assumption to ensure comparability between expansion and non-expansion states before implementation of the Affordable Care Act, and regression models with controls were used to isolate the effects of Medicaid expansion. Households with higher housing burdens were 1.5% more likely to enroll in Medicaid, participation in employer-sponsored insurance enrollment declined by 0.7% and

directly purchased insurance enrollment decreased by 1% among households with higher housing burdens. These effects were more pronounced for individuals above the age of 26.

These studies illustrate how housing market dynamics and affordability shape household behavior. Stutz and Kartman (1982) show that regional demand factors are key drivers of housing price variations. Cao et al. (2024) adds to this by showing that high housing burdens directly influence insurance participation, in the context of public policy. Their findings suggest that increases in housing costs lead households to choose public programs like Medicaid. This behavior underscores the importance of considering housing affordability not just as a housing issue, but an overall determinant of financial and health related decisions.

Comparative Analysis of Findings

Across the reviewed literature, the studies vary in focus, methods, and conclusions, but they collectively show the complex relationship between housing burdens and insurance participation. Through a variety of pathways, Cardon and Hendel (2001) and Chiappori and Salanie (2000) found little evidence of the presence of adverse selection in the health and auto insurance markets, which suggests that price sensitivity may be a more influential factor. In contrast, Cohen (2005) shows that there is adverse selection among experienced auto insurance consumers, which highlights how risk can shape coverage decisions in certain contexts. Outreville (2014) presents empirical evidence that risk preferences and insurance participation are influenced by demographics and behaviors. Cleeton and Zellner (1993) provide a theoretical model that shows that income and risk aversion jointly drive insurance demand. Finally, Stutz and Kartman (1982) and Cao, et al. (2024) connect housing affordability to more financial behaviors, with Cao, et al. (2024) showing that high housing burden households decrease private insurance participation and increase participation with public programs. These studies address

different aspects of the topic, from market structure and individual psychology to policy effects, revealing that insurance decisions under any sort of housing burden are shaped by a mix of financial constraints and behavioral tendencies. My research adds to this foundation by studying how housing burdens specifically affect household participation across multiple types of insurance, including health, auto, life, and personal insurance. By using economic theory, behavioral insights from behavioral economics, and empirical data, my study aims to explain how and why housing costs shape insurance decisions.

Economic Theory

A range of economic theories inform this analysis. From microeconomic theory, concepts such as asymmetric information, budget constraints, and consumer choice are relevant. Moreover, because economics fundamentally examines human behavior, insights from behavioral economics, including decision biases and heuristics, also play a critical role in shaping financial outcomes and household choices.

As Varian (2014) defines, a consumer's budget constraint mandates the amount of money spent on two or more goods or services must be less than or equal to the amount of money that is earned like income. Therefore, household budgets are constrained by the total amount of income that is made. If households are spending a significant portion of their income (30+%) on housing, that constrains their budget for other goods and services. Through including other expenditures in the models, we can see what some of the budget is going to and the level of rationality of these households.

Liberto (2025) states that consumer theory looks at how people or households make purchases based on their limited budgets and preferences and the main assumptions of consumer theory are that the consumers make rational choices; the choice made will have the highest

utility, non-satiation, and decreasing marginal utility. There is something to be said about the relationship between risk and utility. If someone is risk-averse, their utility will be maximized by buying insurance, because they believe that there is value to having insurance, since they feel that they are at risk for an accident of some sort. The converse is also true where the high-risk individuals find more utility in not buying insurance versus having insurance because they feel that they are not at risk of an accident or emergency or gain happiness from the presence of risk.

Not only do the psychological aspects play a role in decision-making, but so does the amount of available information. Asymmetric information is when one party has more information than the other usually concerning a transaction (Bloomenthal, 2025). In this case, the consumer has more information about their risk level than the policy insurer; however, with an increase in technology, risk is easily seen by the insurers. As a result, if insurers do not have access to this personal information, they will struggle with making accurate pricing policies or assessing risks which could lead to losses. This can lead to adverse selection and insufficient coverage options.

Economic theory highlights the fact that there is asymmetric information in insurance markets, where the consumers possess more information about their own risks and needs than the insurers. This imbalance is the same for all types of insurance, where individual level information, like medical history or driving habits, is not fully known by providers. Asymmetric information is present in housing and insurance markets in ways that directly affect affordability for households. In rental markets, landlords don’t know everything about a tenant’s reliability, so they often raise rents or add strict screening requirements, while tenants may not realize the true condition of a property until after moving in, leaving them with hidden costs like high utility bills. In mortgage markets, lenders face uncertainty about borrower risk, which leads to higher

interest rates or tighter credit rules, while buyers themselves may lack full knowledge about neighborhoods or property quality and end up overpaying. Insurance markets mirror these problems: providers can't perfectly assess household risks, so premiums rise, adding to housing costs, while households may underinsure because they don't fully understand policy terms or underestimate their exposure. Together, these gaps in information make housing less affordable, push households toward riskier financial positions, and hit lower-income families hardest. Behavioral economics also suggests that financial stress, like having a housing burden, hinders decision-making by intensifying biases like loss aversion, status quo, overconfidence and confirmation bias (Lyons and Kass-Hanna, 2021). These biases can persuade individuals to undervalue the importance of insurance, delay their enrollment, or choose minimal coverage. For example, life insurance may be less of a priority because it is a long-term policy; health insurance may be avoided even though you know your risk; personal insurance may seem unnecessary; and vehicle insurance coverage may be limited to the legally required minimum. Based on these insights from theory, my hypothesis is that the coefficient for the housing burden variable will be negative for all types of insurance, indicating that higher housing costs relative to income are associated with lower insurance expenditures across all four types, all else equal.

Data

This study uses data from the Bureau of Labor Statistics' Consumer Expenditure Surveys from 2018 Q2 to 2024 Q1. For all the expenditure variables, I use the previous quarter data which allows for more accurate data. This dataset provides detailed data on household-level expenditure data and demographic information. In total, across the twenty-four quarters, there are 91,991 observations. Each insurance type has its own regression model, with the dependent

variable being the log transformation, which is possible by adding one to each the entries, since there are values of 0 (meaning no expenditures) in all the insurance expenditure variables.

Each model has the same demographic factors included: age, ethnicity (race and Hispanic origin), marital status, gender, highest education level between the reference person and their spouse, number of hours week by the reference person and their spouse, family size, housing tenure, and state of residence. Also, a temporal variable will be included that measures the number of quarters since 2018. This is important because there was a recession during the time period of data I looked at. The housing burden variable is the sum of all mandatory expenditures related to housing divided income before taxes. I removed any observations where housing burden that was below 0 and above 1. The categorical variables were put into the model as factor variables to show the group-level differences. The whole table of all of the variable definitions is in Appendix A.

Table 1: Variable Definitions

HOUSB	Housing Burden variable (Sum of Mandatory housing expenditures divided by gross income)
AGE_REF	Age of the reference person
CUTENURE	= 1 if the house is owned with mortgage = 2 if the house is owned without a mortgage = 3 if the house is owned mortgage NR = 4 if the house is rented = 5 if the house is occupied without payment of cash rent = 6 if student housing
FAM_SIZE	Family Size
TINCHRS	Number of hours worked per week by the reference person and their spouse
MARITAL1	= 1 if the reference person is married = 2 if the reference person is widowed = 3 if the reference person is divorced = 4 if the reference person is separated = 5 if the reference person has never been married
REF_RACE	= 1 if the reference is white = 2 if the reference is black = 3 if the reference is native American = 4 if the reference is Asian = 5 if the reference is other = 6 if the reference is multi-race
SEX_REF	= 1 if the reference person is female = 0 if the reference person is male
HISP_REF	= 1 if the reference person is non-Hispanic = 0 if the reference person is Hispanic
HIGH_EDU	= 0 if no education = 10 if completed 8th grade = 11 if finished 9th-12th grade, but no diploma = 12 if high school graduate = 13 if have completed some college, no degree = 14 if completed associate's degree = 15 if completed bachelor's degree = 16 if completed master's or doctorate degree
QSQ118	Quarters since Q1 2018

Each model also has its own specific predictors relating to that type of insurance. For example, vehicle insurance has a variable that measures the number of owned vehicles in the household. As the number of vehicles increases, so does the cost of insurance. Also included in this model are variables that measure expenditures on gas, expenditures on vehicle financing, expenditures on maintenance and repairs, expenditures on vehicle rentals, leases and licenses, expenditures on public and other transportation, and expenditures on vehicle purchases (down payment, principal and interest, purchase amount).

For the life insurance model, the only variables I added were the expenditures on tobacco and alcohol. If these expenditures are higher, one is more likely to have health problems and die sooner. For personal insurance, I did not add any additional variables. There are two inputs into the personal insurance expenditure variable, which are expenditures on life insurance and expenditures on retirement, pensions, and social security. If you add those into the model, r-squared is one which could indicate overfitting.

For health insurance, there are more variables that can be used. The dataset provides data on alcohol and tobacco expenditures, which is useful if someone consumes enough alcohol or tobacco; they will eventually have health problems and need insurance because they cannot afford out-of-pocket expenses. There are some other important indicators like expenditures on medical services, medical supplies, and prescription drugs. These are indicators of someone's health, and it is expected that if these numbers are higher, there is a necessity for insurance to cover those costs. Also, something to consider is that these numbers could be lower because of the insurance, which suggests there is endogeneity in the model. In light of the endogeneity, I use money spent on food.

Table 2: Summary Statistics

Summary Statistics	Mean	SD	Min	Q1	Median	Q3	Max
HOUSB	0.2	0.18	0	0.08	0.15	0.26	1
AGE_REF	52.87	17.53	18	38	53	67	88
CUTENURE	2.3	1.32	1	1	2	4	6
FAM_SIZE	2.4	1.43	1	1	2	3	16
TINCHRS	41.26	32.09	0	0	40	65	170
MARITAL1	2.33	1.6	1	1	1	3	5
REF_RACE	1.4	1.01	1	1	1	1	6
SEX_REF	0.51	0.49	0	0	1	1	1
VEHQ	1.86	1.45	0	1	2	2	24
FOODPQ	1670.63	1521.75	0	693	1253	2167	41406
ALCBEVPQ	101.9	230.08	0	0	0	107	6500
GASMOPQ	380.71	459.38	0	100	240	500	15000
VEHFINPQ	45.64	117.06	0	0	0	34	2289
MAINRPPQ	154.3	514.12	0	0	0	75	35000
VEHINSPQ	270.27	398.78	1	1	102	433	27001
VRNTLOPQ	118.29	523.44	0	0	0	49	42000
PUBTRAPQ	136.85	700.58	0	0	0	0	37130
HLTHINPQ	652.69	808.71	1	101	415	897.5	12065
MEDSRVPQ	188.17	804.97	0	0	0	60	45000
PREDRGPQ	0	0.04	0	0	0	0	1
MEDSUPPQ	24.21	184.18	0	0	0	0	16385
TOBACCPQ	51.67	194.89	0	0	0	0	10400
PERINSPQ	1511.73	2272.3	1	169.04	766	1948.01	63333.46
LIFINSPQ	93.28	444.9	1	1	1	1	40001
RETPENPQ	1418.45	2160.78	0	92.49	716.15	1845.87	63272.46
STATE	27.47	16.44	1	12	27	41	55
EVEHPURP	625.59	3022.08	0	0	0	540	200000
HISP_REF	0.85	0.34	0	1	1	1	1
HIGH_EDU	14.04	1.64	0	13	14	15	16
QSQ118	11.23	6.93	0	5	11	17	23

Over 75% falls under what is considered by the literature to be a high housing burden. More than 50% of the population is above the age of 50, which could explain the lower family sizes, especially if they have children that have already moved out of the house. Out of the housing tenures, it seems like owning the house with a mortgage is the one that appears the most

because at least 25% of the sample fits into the category, which can't necessarily be said about the other categories. Over half of the sample is married, which is surprising given the fact that at least 75% of this sample spends nothing on life insurance (without the adjustment). Most of the sample is white and about half of the sample is female.

Empirical Model

My hypothesis is that housing burden negatively affects insurance spending; meaning households that have higher housing costs allocate less money towards insurance. To test my hypothesis, I estimate separate linear regression models for each type of insurance. The dependent variable in each model is the log of the insurance expenditures (after adding one to each of the observations) for that type of insurance. Each model shares the demographic factors of age, race, Hispanic origin, marital status, gender, highest education level between the reference person and their spouse, number of hours week by the reference person and their spouse, family size, housing tenure, and state of residence. Additionally, a temporal variable is included that measures the number of quarters since 2018, and the housing burden variable is the sum of all mandatory expenditures related to housing divided income before taxes. State-level fixed effects are in the model to account for geographic differences, and each model also has its own specific predictors relating to that type of insurance. Each model follows the same structure:

Vehicle Insurance

$$\log(AdjVEHINSPQ) \;=\; \beta_0 + \beta' X_i + \delta_{iv} + \varepsilon_i$$

Life Insurance

$$\log(AdjLIFINSPQ) \;=\; \beta_0 + \beta' X_i + \delta_{il} + \varepsilon_i$$

Personal Insurance

$$\log(AdjPERINSPQ) = \beta_0 + \beta' X_i + \varepsilon_i$$

Health Insurance

$$\log(AdjHLTHINPQ) \; = \; \beta_0 + \beta' X_i + \delta_{ih} + \varepsilon_i$$

where

- X_i = vector of demographic and household controls, including: HOUSB (housing burden), AGE_REF (age of reference person), CUTENURE (housing tenure), FAM_SIZE (family size), TINCHRS (total number of hours worked by the reference person and their spouse), MARITAL1 (marital status of the reference person), REF_RACE (race of reference person), SEX_REF (sex of reference person), STATE (state fixed effects), HISP_REF (Hispanic origin of reference person), HIGH_EDU (highest education between reference person and spouse), QSQ118 (quarters since 2018 Q1, in terms of expenditure data)
- δ_{iv} = vehicle-specific controls, such as: VEHQ, GASMOPQ, VEHFINPQ, MAINRPPQ, VRNTLOPQ, PUBTRAPQ, EVEHPURP
- δ_{il} = life-insurance-specific controls, such as: ALCBEVPQ, TOBACCPQ
- δ_{ih} = health-specific controls, such as: ALCBEVPQ, FOODPQ, MEDSRVPQ, PREDRGPQ, MEDSUPPQ, TOBACCPQ
- ε_i = error term.

The hypothesis is tested by looking at the sign and statistical significance of coefficient of the housing burden variable and the fit of the model is assessed using the R-squared and the F-statistic. By comparing the results between insurance types, we can see if the results are the same across the board and what that means for housing costs and insurance expenditures.

Results

Table 3: Coefficients and significance

	Vehicle Insurance	Life Insurance	Personal Insurance	Health Insurance
(Intercept)	1.165*** (0.310)	-0.008 (0.242)	3.182*** (0.209)	1.547*** (0.253)
Housing Burden	2.127*** (0.056)	0.482*** (0.044)	-0.742*** (0.037)	-0.538*** (0.046)
Age of reference person	-0.002** (0.001)	0.015*** (0.001)	-0.021*** (0.001)	0.052*** (0.001)
Owned without mortgage	0.022 (0.027)	-0.297*** (0.021)	-0.415*** (0.018)	-0.261*** (0.022)
Rented	-0.199*** (0.026)	-0.492*** (0.020)	-0.497*** (0.017)	-0.678*** (0.021)
Occupied without payment of cash rent	-0.250** (0.088)	-0.513*** (0.069)	-0.994*** (0.059)	-1.317*** (0.072)
Student housing	-1.058*** (0.230)	-0.361* (0.180)	-2.332*** (0.155)	-2.355*** (0.188)
Family size	0.002 (0.008)	0.067*** (0.006)	0.196*** (0.005)	-0.019** (0.007)
Number of hours worked per week	0.004*** (0.000)	0.007*** (0.000)	0.048*** (0.000)	0.000 (0.000)
Widowed	0.081* (0.039)	-0.264*** (0.031)	0.116*** (0.026)	-0.300*** (0.032)
Divorced	0.164*** (0.031)	-0.257*** (0.025)	0.713*** (0.021)	-0.504*** (0.026)
Separated	-0.128+ (0.068)	-0.345*** (0.054)	0.821*** (0.046)	-0.825*** (0.056)
Never married	-0.003 (0.033)	-0.358*** (0.025)	0.792*** (0.022)	-0.477*** (0.026)
Black	-0.039 (0.033)	0.388*** (0.026)	0.095*** (0.022)	-0.141*** (0.027)
Native American	0.037 (0.133)	0.272** (0.104)	-0.139 (0.090)	-0.235* (0.109)
Asian or Pacific Islander	-0.211*** (0.042)	-0.155*** (0.033)	0.162*** (0.028)	-0.036 (0.034)
Other	0.049 (0.134)	0.053 (0.105)	0.227* (0.090)	0.161 (0.109)
Multirace	-0.045 (0.074)	0.216*** (0.058)	0.092+ (0.050)	0.008 (0.061)
Female	-0.002 (0.019)	0.108*** (0.015)	-0.090*** (0.013)	0.068*** (0.016)

Non-Hispanic	-0.040 (0.031)	0.261*** (0.024)	-0.172*** (0.021)	0.657*** (0.025)
Completed 8th grade	-0.130 (0.308)	-0.014 (0.241)	-0.044 (0.208)	-0.046 (0.252)
Completed 12th grade	0.126 (0.298)	0.132 (0.234)	0.368+ (0.201)	-0.513* (0.244)
Highschool graduate	0.423 (0.295)	0.186 (0.231)	0.830*** (0.199)	0.059 (0.241)
Some college, no degree	0.587* (0.295)	0.322 (0.231)	1.085*** (0.199)	0.306 (0.241)
Associate's degree	0.627* (0.296)	0.468* (0.232)	1.358*** (0.200)	0.599* (0.242)
Bachelor's degree	0.605* (0.295)	0.450+ (0.231)	1.563*** (0.199)	0.836*** (0.241)
Master's or doctorate degree	0.667* (0.296)	0.686** (0.231)	1.707*** (0.199)	0.985*** (0.241)
Number of vehicles	0.183*** (0.008)			
Gas and motor oil expenditures	0.001*** (0.000)			
Vehicle financing expenditures	0.002*** (0.000)			
Maintenance and repair expenditures	0.000*** (0.000)			
Vehicle rentals, leases and licenses expenditures	0.000*** (0.000)			
Public transportation expenditures	-0.000 (0.000)			
Vehicle purchases expenditures	0.000*** (0.000)			
Alcohol expenditures		0.001*** (0.000)		0.000*** (0.000)
Tobacco expenditures		0.000 (0.000)		-0.000 (0.000)
Food expenditures				0.000*** (0.000)
Medical services expenditures				0.000*** (0.000)

Prescription drugs expenditures				-0.054 (0.187)
Medical supplies expenditures				0.000*** (0.000)
State-fixed effects	Yes	Yes	Yes	Yes
Number of observations	91991	91991	91991	91991
R-squared	0.127	0.083	0.542	0.248
F	180.17	119.88	1624.27	415.25
p-value of F-statistic	<2.2e-16***	<2.2e-16***	<2.2e-16***	<2.2e-16***
Significance	+p < 0.1	* p < 0.05	** p < 0.01	*** p < 0.001

For the full results of the state-fixed effects see Appendix B. The reference groups in this study are individuals who are married, individuals with no education, individuals who own a home with a mortgage, and individuals who are white. Relative to the reference groups, vehicle insurance expenditures increase along with housing burden, which means a 0.01 or 1% increase in housing burden corresponds to a 2.127% increase in vehicle insurance spending. Economically, this suggests that households with higher housing burdens may own higher cost vehicles or are high risk, which both increases how much is paid for insurance. Additional vehicles increase expenditures by 18.3% (0.183), and related costs like gas, financing, maintenance, rentals and licenses, and purchases are all positive and significant with small effects but is consistent with the idea that greater vehicle usage and investment increase risk. Compared to those who own a house and have a mortgage, renters (-0.199), cash-rent households (-0.250), and student housing residents (-1.058) spend less, with student housing residents spending the least by 105.8% which is because people in this situation likely don't own a vehicle, own fewer vehicles, or own lower-value vehicle and drive less overall. Relative to married households, divorced individuals spend more (0.164), while separated individuals spend

less (-0.128) which reflects how changes in family dynamics impact driving patterns. Education increases vehicle insurance expenditures: bachelor's (0.605) and graduate degrees (0.667) are related to increased spending, this could mean that because they have more education, they get better jobs and buy more expensive cars, or they are more risk-averse and buy the maximum protection. The effects for race are mixed, compared to the references groups Asians spend about 21% (-0.211) less while other races are not significant. The model explains 12.7% of the variation.

Relative to the reference groups, life insurance expenditures increase with housing burden (0.482), meaning that a 0.01 increase in housing burden corresponds to a 0.48% increase in life insurance spending. This suggests that households with higher housing burdens may invest in additional protection against unexpected changes in income. Age also has a strong and positive effect (0.015), which is consistent with the idea that as people age insurance demand increases as does mortality risk. Compared to owners who have a mortgage, renters, cash-rent, households, and student housing residents spend less, which is consistent with lower wealth, less dependents, and less of a need for long-term financial protection. Relative to married households, widowed, divorced, separated, and never married households all spend less, which indicates fewer household responsibilities and fewer beneficiaries. Education increases expenditures; associates and graduate degrees are correlated with higher spending, likely due to higher overall income and financial literacy. Females and non-Hispanic individuals spend more, while the other race effects are mixed, Black and Native American households spend more, while Asian households spend less, which could reflect cultural differences. Alcohol expenditures are positively correlated with life insurance spending (0.001), which captures behavioral risk. The model explains 8.3% of the variation.

Relative to the reference groups, personal insurance expenditures decrease when housing burden increases (-0.742), meaning households with higher housing costs relative to income spend less on supplementary insurance, which confirms my hypothesis. Economically, this suggests that when budgets are tighter, households prioritize necessary policies like life or health insurance. Age also decreases spending (-0.021), which aligns with older households focusing on core insurance necessities. Education has a strong and positive effect, bachelor's and graduate degrees vastly increase expenditures, which reflects higher incomes and better awareness of supplementary coverage. Divorced, separated, never married, and widowed individuals spend more than married individuals, consistent with single adults increasing protection because they lack a second income. Family size also increases expenditures (0.196), likely due to increased demand for broader coverage. The effects for race are mixed: Asian and Black individuals spend more while Native Americans spend less. Females and non-Hispanic individuals spend less, which suggests differences in risk preferences. Alcohol expenditures are positively associated with personal insurance spending. The model explains a significant share of the variation (54.2%).

Health insurance expenditures decrease when housing burden increases (-0.538), suggesting that tighter budgets lessen coverage or choose more affordable coverage. Age increases spending (0.052), which reflects that a person's health deteriorates as a person ages. Renters, cash-rent households, and student housing residents spend less than owners, consistent with lower income and limited access to employer sponsored insurance. All non-married individuals spend less than married households, which is likely due to fewer dependents and fewer employer benefits. Education increases spending; bachelor's and graduate degrees are linked to higher amounts of coverage which reflects higher income and more access to employer

sponsored plans. Non-Hispanic households and females spend marginally more, while Black and Asian households spend less. Family size reduces spending, and medical services and supplies are associated with higher insurance spending. The model explains 24.8% of the variation.

Conclusion

This study investigates the relationship between a household's housing burden and their insurance expenditures for four types of insurance: vehicle, life, personal, and health insurance. Housing burden is positively correlated with expenditures for vehicle, and life insurance, contradicting the hypothesis. In the personal and health insurance models, housing burden did have a negative and significant impact, which is consistent with my hypothesis. In all of the models, the demographic and socioeconomic factors like age, number of hours worked, education, family size, and marital status were significant predictors of insurance expenditures, while state-level fixed effects indicated that there are substantial geographic differences.

These findings give nuanced contributions to the prior literature on insurance demand and expenditures, financial behavior, and housing affordability. The negative relationship between housing burden and personal and health insurance expenditures lines up with prior studies on financial constraints and the substitution effect in budgeting (Stutz & Kartman, 1982; Cao et al., 2024). It also supports the idea that higher housing costs can decrease participation in optional financial products/assets, not just insurance, as suggested by Cleeton and Zellner (1993) and Lyons and Kass-Hanna (2021). On the other hand, the positive relationships that were observed for vehicle and life insurance challenge this narrative and suggest that housing burden also serves as a proxy for financial activity, which is consistent with consumer theory (Liberto, 2025) and behavioral models (Outreville, 2014; Varian 2014). The role of education and number of hours worked per week by the reference person and their spouse in predicting insurance expenditures

reflects findings from Chiappori and Salanie (2000) and Cohen (2005), who emphasize the significance of observable traits in insurance participation. Moreover, the racial disparities observed in the spending patterns relate to the literature on asymmetric information and access to insurance markets (Cardon & Hendel, 2001; Bloomenthal, 2025).

A strength of this study is the use of a large, and representative dataset with detailed information on expenditure and demographics, which allows for precise estimation and insurance specific models. Including the state variable and insurance specific controls enhances the robustness of the analysis. However, the housing burden variable may be endogenous to broader financial behavior and variables not in the model. While the models capture variation across insurance types, they do not capture behavioral biases or preferences that influence decisions.

These findings suggest that housing burden does not decrease insurance spending overall, but its effects depend on the type of insurance. Policymakers who are concerned with underinsurance should think about targeted policies for personal and health insurance markets, where the financial burden decreases coverage. Expanding access to affordable insurance products, especially renters and low-income families, could help decrease risk exposure. The role of education in predicting insurance expenditures underscores the necessity for financial literacy initiatives.

References

Bloomenthal, A. (2025, July 31). *Asymmetric information in economics explained*. Investopedia. https://www.investopedia.com/terms/a/asymmetricinformation.asp

Cao, Y., Su, Y., Wang, G., & Zhang, C. (2024). Health insurance coverage changes under the Affordable Care Act among High Housing Cost Households, 2010–18. *Health Economics*. https://doi.org/10.1002/hec.4912

Cardon, J. H., & Hendel, I. (2001). Asymmetric Information in Health Insurance: Evidence from the National Medical Expenditure Survey. *The RAND Journal of Economics*, *32*(3), 408. https://doi.org/10.2307/2696362

Chiappori, P., & Salanie, B. (2000). Testing for asymmetric information in insurance markets. *Journal of Political Economy*, *108*(1), 56–78. https://doi.org/10.1086/262111

Cleeton, D. L., & Zellner, B. B. (1993). Income, risk aversion, and the demand for insurance. *Southern Economic Journal*, *60*(1), 146. https://doi.org/10.2307/1059939

Cohen, A. (2005). Asymmetric Information and Learning: Evidence from the Automobile Insurance Market. *The Review of Economics and Statistics*, *87*(2), 197–207. https://doi.org/10.1162/0034653053970294

Krstic, B., & Krstic, M. (2015). Rational choice theory and random behaviour. *Ekonomika*, *61*(1), 1–13. https://doi.org/10.5937/ekonomika1501001k

Liberto, D. (2025, August 21). *Consumer theory explained: definition, goals, and Real-World applications*. Investopedia. https://www.investopedia.com/terms/c/consumer-theory.asp

Lyons, A., & Kass-Hanna, J. (2021). Behavioral economics and financial decision making. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3887605

Outreville, J. F. (2014). Risk Aversion, Risk Behavior, and Demand for Insurance: a survey. *Journal of Insurance Issues*, *37*(2), 158–186. https://econpapers.repec.org/RePEc:wri:journl:v:37:y:2014:i:2:p:158-186

Stutz, F. P., & Kartman, A. E. (1982). Housing affordability and spatial price variations in the United States. *Economic Geography*, *58*(3), 221. https://doi.org/10.2307/143511

Varian, H. R. (2014). *Intermediate Microeconomics with Calculus*. https://ci.nii.ac.jp/ncid/BB1838634X

Appendix A: Variable Definitions

HOUSB	Housing Burden variable (Sum of Mandatory housing expenditures divided by gross income)
AGE_REF	Age of the reference person
CUTENURE	= 1 if the house is owned with mortgage = 2 if the house is owned without a mortgage = 3 if the house is owned mortgage NR = 4 if the house is rented = 5 if the house is occupied without payment of cash rent = 6 if student housing
FAM_SIZE	Family Size
TINCHRS	Number of hours worked per week by the reference person and their spouse
MARITAL1	= 1 if the reference person is married = 2 if the reference person is widowed = 3 if the reference person is divorced = 4 if the reference person is separated = 5 if the reference person has never been married
REF_RACE	= 1 if the reference is white = 2 if the reference is black = 3 if the reference is native American = 4 if the reference is Asian = 5 if the reference is other = 6 if the reference is multi-race
SEX_REF	= 1 if the reference person is female = 0 if the reference person is male
HISP_REF	= 1 if the reference person is non-Hispanic = 0 if the reference person is Hispanic
HIGH_EDU	= 0 if no education = 10 if completed 8th grade = 11 if finished 9th-12th grade, but no diploma = 12 if high school graduate = 13 if have completed some college, no degree = 14 if completed associate's degree = 15 if completed bachelor's degree = 16 if completed master's or doctorate degree
QSQ118	Quarters since Q1 2018
VEHINSPQ	Vehicle insurance expenditures in the previous quarter

VEHQ	Number of owned vehicles
GASMOPQ	Gas and motor oil expenditures in the previous quarter
VEHFINPQ	Vehicle finance charges in the previous quarter
MAINRPPQ	Expenditures on Maintenance on Repairs in the previous quarter
VRNTLOPQ	Expenditures on vehicle rentals, leases and licenses in previous quarter
PUBTRAPQ	Expenditures on public transportation in the previous quarter
EVEHPURP	Expenditures on vehicle purchases
LIFINSPQ	Expenditures on life insurance in the previous quarter
ALCBEVPQ	Expenditures of alcoholic beverages in the previous quarter
TOBACCPQ	Expenditures on Tobacco in the previous quarter
PERINSPQ	Expenditures on personal insurance (life insurance, retirement, and pensions) in the previous quarter
HLTHINPQ	Expenditures on health insurance in the previous quarter
FOODPQ	Expenditures on food in the previous quarter
MEDSRVPQ	Expenditures on medical services in the previous quarter
PREDRGPQ	Expenditures on prescription drugs in the previous quarter
MEDSUPPQ	Expenditures on Medical Supplies in the previous quarter

Appendix B: Complete results

	Vehicle Insurance	Life Insurance	Personal Insurance	Health Insurance
(Intercept)	1.165*** (0.310)	-0.008 (0.242)	3.182*** (0.209)	1.547*** (0.253)
Housing Burden	2.127*** (0.056)	0.482*** (0.044)	-0.742*** (0.037)	-0.538*** (0.046)
Age of reference person	-0.002** (0.001)	0.015*** (0.001)	-0.021*** (0.001)	0.052*** (0.001)
Owned without mortgage	0.022 (0.027)	-0.297*** (0.021)	-0.415*** (0.018)	-0.261*** (0.022)
Rented	-0.199*** (0.026)	-0.492*** (0.020)	-0.497*** (0.017)	-0.678*** (0.021)
Occupied without payment of cash rent	-0.250** (0.088)	-0.513*** (0.069)	-0.994*** (0.059)	-1.317*** (0.072)
Student housing	-1.058*** (0.230)	-0.361* (0.180)	-2.332*** (0.155)	-2.355*** (0.188)
Family size	0.002 (0.008)	0.067*** (0.006)	0.196*** (0.005)	-0.019** (0.007)
Number of hours worked per week	0.004*** (0.000)	0.007*** (0.000)	0.048*** (0.000)	0.000 (0.000)
Widowed	0.081* (0.039)	-0.264*** (0.031)	0.116*** (0.026)	-0.300*** (0.032)
Divorced	0.164*** (0.031)	-0.257*** (0.025)	0.713*** (0.021)	-0.504*** (0.026)
Separated	-0.128+ (0.068)	-0.345*** (0.054)	0.821*** (0.046)	-0.825*** (0.056)
Never married	-0.003 (0.033)	-0.358*** (0.025)	0.792*** (0.022)	-0.477*** (0.026)
Black	-0.039 (0.033)	0.388*** (0.026)	0.095*** (0.022)	-0.141*** (0.027)
Native American	0.037 (0.133)	0.272** (0.104)	-0.139 (0.090)	-0.235* (0.109)
Asian or Pacific Islander	-0.211*** (0.042)	-0.155*** (0.033)	0.162*** (0.028)	-0.036 (0.034)
Other	0.049 (0.134)	0.053 (0.105)	0.227* (0.090)	0.161 (0.109)
Multirace	-0.045 (0.074)	0.216*** (0.058)	0.092+ (0.050)	0.008 (0.061)
Female	-0.002 (0.019)	0.108*** (0.015)	-0.090*** (0.013)	0.068*** (0.016)
Non-Hispanic	-0.040 (0.031)	0.261*** (0.024)	-0.172*** (0.021)	0.657*** (0.025)

Completed 8th grade	-0.130 (0.308)	-0.014 (0.241)	-0.044 (0.208)	-0.046 (0.252)
Completed 12th grade	0.126 (0.298)	0.132 (0.234)	0.368+ (0.201)	-0.513* (0.244)
Highschool graduate	0.423 (0.295)	0.186 (0.231)	0.830*** (0.199)	0.059 (0.241)
Some college, no degree	0.587* (0.295)	0.322 (0.231)	1.085*** (0.199)	0.306 (0.241)
Associate's degree	0.627* (0.296)	0.468* (0.232)	1.358*** (0.200)	0.599* (0.242)
Bachelor's degree	0.605* (0.295)	0.450+ (0.231)	1.563*** (0.199)	0.836*** (0.241)
Master's or doctorate degree	0.667* (0.296)	0.686** (0.231)	1.707*** (0.199)	0.985*** (0.241)
Alaska	-0.134 (0.094)	-0.608*** (0.074)	0.286*** (0.063)	-0.213** (0.077)
Arizona	-0.029 (0.089)	-0.680*** (0.069)	-0.115+ (0.060)	-0.267*** (0.072)
California	-0.158* (0.069)	-0.584*** (0.054)	0.257*** (0.046)	-0.430*** (0.056)
Colorado	0.314*** (0.095)	-0.766*** (0.075)	0.315*** (0.064)	-0.040 (0.078)
Connecticut	0.107 (0.089)	-0.337*** (0.069)	0.191** (0.060)	0.157* (0.072)
Delaware	0.756** (0.252)	-0.461* (0.197)	0.577*** (0.170)	-0.467* (0.205)
District of Columbia	-0.339+ (0.199)	-0.142 (0.156)	0.334* (0.134)	-0.063 (0.163)
Florida	-0.048 (0.073)	-0.772*** (0.057)	-0.110* (0.049)	-0.040 (0.060)
Georgia	0.167+ (0.092)	-0.426*** (0.072)	0.143* (0.062)	0.233** (0.075)
Hawaii	-0.077 (0.097)	-0.743*** (0.076)	0.289*** (0.066)	-0.626*** (0.079)
Illinois	-0.234** (0.083)	-0.408*** (0.065)	0.121* (0.056)	0.145* (0.067)
Indiana	-0.902*** (0.150)	-0.893*** (0.117)	-0.012 (0.101)	-0.271* (0.122)
Iowa	-0.603* (0.247)	-0.381* (0.194)	0.043 (0.167)	-0.227 (0.202)
Kansas	-0.112 (0.108)	-0.071 (0.085)	-0.034 (0.073)	-0.061 (0.088)
Kentucky	0.215* (0.103)	-0.281*** (0.081)	0.047 (0.069)	-0.157+ (0.084)
Louisiana	0.245* (0.099)	-0.928*** (0.077)	-0.192** (0.066)	-0.386*** (0.081)

Maryland	0.113 (0.089)	-0.420*** (0.070)	0.411*** (0.060)	0.196** (0.073)
Massachusetts	-0.213** (0.082)	-0.460*** (0.064)	0.157** (0.055)	0.108 (0.067)
Michigan	-0.110 (0.083)	-0.339*** (0.065)	0.009 (0.056)	-0.302*** (0.068)
Minnesota	-0.143 (0.092)	-0.240*** (0.072)	0.138* (0.062)	0.160* (0.075)
Mississippi	-0.213 (0.138)	0.005 (0.108)	-0.183* (0.093)	-0.011 (0.113)
Missouri	-0.064 (0.099)	-0.026 (0.078)	0.144* (0.067)	0.156+ (0.081)
Nebraska	0.119 (0.088)	-0.249*** (0.069)	0.018 (0.060)	0.151* (0.072)
Nevada	0.192+ (0.107)	-0.352*** (0.084)	0.133+ (0.072)	-0.354*** (0.087)
New Hampshire	-0.085 (0.221)	0.194 (0.173)	0.360* (0.149)	0.404* (0.180)
New Jersey	-0.065 (0.090)	-0.435*** (0.070)	0.250*** (0.060)	-0.190** (0.073)
New York	-0.550*** (0.074)	-0.625*** (0.057)	0.164*** (0.049)	-0.523*** (0.060)
North Carolina	0.001 (0.075)	-0.469*** (0.059)	0.004 (0.051)	-0.084 (0.061)
Ohio	-0.028 (0.077)	-0.384*** (0.061)	0.150** (0.052)	-0.004 (0.063)
Oklahoma	0.272* (0.122)	-0.067 (0.096)	0.273*** (0.083)	-0.078 (0.100)
Oregon	0.039 (0.092)	-0.475*** (0.072)	0.175** (0.062)	-0.314*** (0.075)
Pennsylvania	-0.011 (0.079)	-0.405*** (0.062)	0.129* (0.053)	-0.008 (0.064)
South Carolina	0.110 (0.101)	-0.239** (0.079)	-0.096 (0.068)	0.020 (0.082)
South Dakota	-0.229 (0.161)	-0.740*** (0.126)	-0.230* (0.109)	0.111 (0.132)
Tennessee	-0.007 (0.109)	0.148+ (0.086)	0.140+ (0.074)	-0.039 (0.089)
Texas	0.138+ (0.073)	-0.524*** (0.057)	0.112* (0.049)	-0.036 (0.060)
Utah	0.072 (0.082)	-0.520*** (0.064)	0.029 (0.055)	-0.279*** (0.067)
Virginia	-0.183* (0.088)	-0.241*** (0.069)	0.280*** (0.059)	0.121+ (0.072)
Washington	-0.144 (0.088)	-0.442*** (0.069)	0.371*** (0.059)	-0.520*** (0.072)

West Virginia	-0.215 (0.152)	-0.137 (0.119)	0.119 (0.103)	-0.251* (0.124)
Wisconsin	-0.187* (0.083)	-0.211** (0.065)	0.121* (0.056)	0.438*** (0.068)
Number of vehicles	0.183*** (0.008)			
Gas and motor oil expenditures	0.001*** (0.000)			
Vehicle financing expenditures	0.002*** (0.000)			
Maintenance and repair expenditures	0.000*** (0.000)			
Vehicle rentals, leases and licenses expenditures	0.000*** (0.000)			
Public transportation expenditures	-0.000 (0.000)			
Vehicle purchases expenditures	0.000*** (0.000)			
Alcohol expenditures		0.001*** (0.000)		0.000*** (0.000)
Tobacco expenditures		0.000 (0.000)		-0.000 (0.000)
Food expenditures				0.000*** (0.000)
Medical services expenditures				0.000*** (0.000)
Prescription drugs expenditures				-0.054 (0.187)
Medical supplies expenditures				0.000*** (0.000)
Number of observations	91991	91991	91991	91991
R-squared	0.127	0.083	0.542	0.248
F	180.17	119.88	1624.27	415.25
p-value of F-statistic	<2.2e-16***	<2.2e-16***	<2.2e-16***	<2.2e-16***
Significance	+p < 0.1	* p < 0.05	** p < 0.01	*** p < 0.001